

RIDGETOWN, ON, Canada

WMO: 713070

Lat: 42.454N

Lon: 81.886W

Elev: 206

StdP: 98.88

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-16.0	-22.7	0.5	-18.8	-19.2	0.7	-15.7	11.4	-2.5	10.5	-2.9	2.9	230	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.4	23.4	29.0	22.5	27.5	21.6	24.8	28.7	23.8	27.4	22.9	26.2	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.9	27.0	22.7	17.8	26.0	21.8	16.9	25.1	76.4	29.0	72.4	27.4	68.8	26.2	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.8	7.9	-22.8	33.0	4.1	1.2	-25.8	33.9	-28.2	34.6	-30.5	35.3	-33.5	36.2		
(5)			-23.1	26.4	4.0	1.4	-25.9	27.4	-28.3	28.3	-30.5	29.1	-33.4	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-4.6	-4.1	1.0	7.4	14.0	19.0	21.2	20.2	17.1	10.8	4.5	-0.9		
	DBStd	10.38	6.59	6.22	5.94	4.71	4.78	3.63	3.14	3.02	4.03	4.86	4.94	5.30		
	HDD10.0	1816	452	397	284	106	14	0	0	0	1	49	175	338		
	HDD18.3	3734	711	629	537	328	152	34	9	16	71	238	414	597		
	CDD10.0	1407	0	1	6	28	138	271	348	317	213	75	11	0		
	CDD18.3	286	0	0	0	1	18	55	99	74	33	6	0	0		
	CDH23.3	2178	0	0	0	7	148	414	824	529	230	27	0	0		
	CDH26.7	491	0	0	0	0	30	107	207	94	48	4	0	0		
(14) Wind	WSAvg	3.6	4.4	4.2	4.1	4.4	3.7	3.2	2.6	2.4	2.7	3.4	4.0	4.2		
(15) Precipitation	PrecAvg	917	75	62	61	76	84	79	85	74	81	77	81	75		
	PrecMax	1200	118	125	123	125	152	158	164	133	214	154	148	147		
	PrecMin	720	30	17	22	26	23	31	31	13	18	20	30	31		
	PrecStd	142	22	27	29	26	33	33	29	28	45	32	33	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	13.9	19.5	24.1	29.7	31.7	32.5	31.2	30.6	26.4	18.5	13.3		
		MCWB	11.1	11.4	15.2	15.8	22.0	23.8	24.3	24.4	23.2	21.1	14.3	11.7		
	2%	DB	9.4	9.2	15.0	20.8	27.1	29.7	30.6	29.0	28.0	22.8	15.9	10.3		
		MCWB	8.7	7.4	12.1	14.4	19.8	22.4	23.6	23.4	21.5	18.2	13.0	9.1		
	5%	DB	6.3	5.9	12.2	17.9	24.8	27.6	29.0	27.5	26.0	20.7	13.9	8.0		
		MCWB	5.2	4.3	9.9	12.8	18.7	21.3	22.8	22.2	20.5	17.2	11.4	6.7		
	10%	DB	4.0	3.3	9.5	15.2	22.0	25.7	27.4	26.3	24.1	18.9	12.2	5.8		
		MCWB	3.2	2.0	7.2	10.9	17.0	20.0	21.7	21.4	19.7	15.7	10.1	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.6	15.9	17.5	23.1	25.3	26.5	25.8	24.3	21.9	15.7	11.7		
		MCDB	11.9	13.5	19.3	22.2	27.5	30.0	30.0	28.3	28.9	26.1	17.1	12.9		
	2%	WB	8.8	7.8	12.7	15.4	21.1	23.7	25.0	24.5	22.8	19.7	13.6	9.4		
		MCDB	9.4	9.2	14.3	19.3	25.6	27.7	28.9	27.6	26.3	21.6	15.4	10.2		
	5%	WB	5.5	4.3	10.0	13.6	19.5	22.5	23.9	23.4	21.7	17.7	11.8	6.8		
		MCDB	6.2	5.4	11.8	17.3	23.4	26.1	27.5	26.3	24.5	19.9	13.5	7.7		
	10%	WB	3.1	2.1	7.4	11.5	17.7	21.2	22.9	22.4	20.6	15.9	10.3	4.8		
		MCDB	3.9	3.2	9.2	14.7	21.0	24.6	26.3	25.1	23.1	18.4	12.0	5.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.8	8.2	10.0	10.5	10.3	10.6	10.5	10.7	9.4	7.7	6.2		
		MCDBR	8.1	9.9	11.5	13.9	12.5	11.8	11.3	10.9	11.6	10.7	9.5	7.6		
	5% WB	MCWBR	7.8	8.8	8.9	9.2	7.2	6.5	5.9	6.2	6.3	6.6	7.4	7.1		
		MCWBR	8.0	9.3	10.7	12.2	11.3	10.6	9.9	9.6	10.0	9.9	8.8	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.324	0.344	0.389	0.415	0.436	0.428	0.409	0.377	0.362	0.323	0.300		
	taud		2.393	2.307	2.340	2.271	2.248	2.225	2.271	2.309	2.386	2.403	2.471	2.468		
	Ebn at Noon		838	872	898	878	861	841	843	849	853	820	808	812		
	Edh at Noon		79	101	111	129	135	139	131	123	106	92	74	68		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.07	3.48	4.62	5.52	6.26	6.20	5.42	4.32	2.75	1.68	1.16		
	RadStd		0.14	0.29	0.36	0.42	0.48	0.40	0.37	0.25	0.37	0.23	0.23	0.09		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page